After examining so young a surface covered with congealed magma, you might wonder if there are any active volcanos left. None has been found for certain, but there are a few—for example, one called Maat Mons—that appear to be surrounded by fresh lava and which may indeed still be churning and belching. There is some evidence that the abundance of sulfur compounds in the high atmosphere varies with time, as if volcanos at the surface were episodically injecting these materials into the atmosphere. When the volcanos are quiescent, the sulfur compounds simply fall out of the air. There's also disputed evidence of lightning playing around the mountaintops of Venus, as sometimes happens on active volcanos on Earth. But we do not know for certain whether there is ongoing vulcanism on Venus. That's a matter for future missions.

Some scientists believe that until about 500 million years ago the Venus surface was almost entirely devoid of landforms. Streams and oceans of molten rock were relentlessly pouring out of the interior, filling in and covering over any relief that had managed to form. Had you plummeted down through the clouds in that long-ago time, the surface would have been nearly uniform and featureless. At night the landscape would have been hellishly glowing from the red heat of molten lava. In this view, the great internal heat engine of Venus, which supplied copious amounts of magma to the surface until about 500 million years ago, has now turned off. The planetary heat engine has finally run down.

In another provocative theoretical model, this one by the geophysicist Donald Turcotte, Venus has plate tectonics like the Earth's—but it turns off and on. Right now, he proposes, the plate tectonics are off; "continents" do not move along the surface, do not crash into one another, do not thereby raise mountain ranges, and are not later subducted into the deep interior. After hundreds of millions of years of quiescence, though, plate tectonics always breaks out and surface features are flooded by lava, destroyed by mountain building, subducted, and otherwise obliterated. The last such breakout ended about 500 million years ago, Turcotte suggests, and everything has been quiet since. However, the presence of coronae may signify—on timescales that are geologically in the near